



Infection with a human-derived enteroinvasive *Escherichia coli* strain altered intestinal barrier function in guinea pigs

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Abstract

Background/aims The aim was to characterize a bacterium causing intestinal mucosal barrier damage and to identify the possible invasion mechanism.

Materials and methods The intestinal permeability and tight junction protein levels were detected in guinea pigs infected with *Escherichia coli* D-09 via immunofluorescence analysis and western blotting. In order to explain this invasion mechanism at the gene level, whole genome sequencing analysis was performed on this bacterium.

Results The results showed an increased intestinal permeability and upregulated expression of the leaky protein claudin-2 in both the colon and liver of the infected animals. In addition, the draft genome of *E. coli* D-09 comprised 42 scaffolds (size, > 645 bp) with a total size of 4,679,567 bp. A total of 4379 protein coding genes were identified, which contained 45 antibiotic resistance and 86 virulence-related genes and covered 88.0% of the whole genome.

Conclusions This study verified that the human-derived enteroinvasive *E. coli* strain could destroy intestinal barrier function in guinea pigs. Additionally, our data first characterized the genome features of *E. coli* O124:K72 D-09, which may provide new insights into the possible invasion mechanism.

Keywords Whole genome sequencing · *E. coli* O124:K72 D-09 · Intestinal barrier function · Acute cholecystitis

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Introduction

Bacterial infection is one of the important factors that cause acute cholecystitis. Enterobacteriaceae, especially *Escherichia coli*, were the most common microorganisms that were isolated from the bloodstream or bile of acute cholecystitis patients (Wang et al. 2003; Yang et al. 2019). An *E. coli* strain, D-09, was also isolated from the bile of patients with acute cholecystitis (Liu et al. 2015). Serological tests identified this strain as *E. coli* O124:K72, which belongs to enteroinvasive *Escherichia coli* (EIEC). EIEC is considered as one of the principal etiologic agent of bacillary dysentery and continues to pose a threat to public health. EIEC is gram-negative bacteria responsible for bacillary dysentery in humans, which is characterized by invasion and inflammatory destruction of the human colonic epithelium (Belotserkovsky and Sansonetti 2018). Since 2012, EIEC serotype O96:H19 has been recognized in Europe, Colombia, and most recently Uruguay (Iqbal et al. 2022).

EIEC serotype O124:K72 was first reported in 1963 as a pathogen associated with diarrhea (Sereny 1963). This strain was firstly discovered in the bile of patients with acute cholecystitis (Sereny 1963). With the advent of next-generation sequencing technology, bacterial whole-genome sequencing has positioned this technology as the most effective tool for species identification, antimicrobial susceptibility prediction and resistance, virulence determinant detection, and outbreak detection (Didelot et al. 2012; Fleece et al. 2018). Mild inflammation and morphological changes were observed in the gallbladder of guinea pigs after infection with *E. coli* O124:K72 D-09 (Liu et al. 2015). The aim of this work is to assess *E. coli* O124:K72 D-09 causing intestinal mucosal barrier damage and to verify the protein levels of claudin-2 and occludin by immunofluorescence and western blotting and identify the possible invasion mechanism via bacterial whole-genome sequencing.

Materials and methods

Isolation and identification of bacterial strain

E. coli O124:K72 D-09 was isolated from the bile of a patient with acute cholecystitis who was diagnosed at the First Affiliated Hospitals of Dalian Medical University of China. Biliary decompression and drainage was performed by percutaneous transhepatic gallbladder drainage surgery. Bile from the drainage was collected aseptically and stored in sterile plastic sample bottles on ice and processed

within 2 h. For the bacterial isolation, as described in our previous article [Liu et al. 2015], briefly, 500 µl bile was added into 50 ml of sterile Luria–Bertani liquid medium, and the culture was incubated for 14 to 18 h at 37 °C. The culture was then spread out on Luria–Bertani solid agar to separate single colonies. To validate this strain, as described in our previous article [Liu et al. 2015], we adopted full-length 16S rRNA gene sequencing and mass spectrometry (Ultraflex extreme MALDI-TOF/TOF MS) analysis respectively for bacterial identification. And their returned results were consistent—*E. coli* (Please refer to Fig. S1a and Fig. S1b of Liu's paper for identification results). Identified isolate was inoculated in LB liquid medium containing 20% glycerol and stored at –80 °C for further study. The serotyping assay was carried out as reported (Liu et al. 2015). Serotest *E. coli* (O&K) Antisera (S&A Reagents Lab Ltd., Part, Thailand) was utilized for serotype identification.

Minimal inhibitory concentration testing

Antibiotic sensitivity analysis of *E. coli* O124:K72 D-09 was completed using freeze-dried bacterial quantitative drug sensitivity (MIC: minimal inhibitory concentration) test kit (Jin Zhang Technology Development Co., Ltd, China). Nine antibiotics were tested: tetracycline, ciprofloxacin, nalidixic acid, ceftiofur, cefotaxime, gentamicin, trimethoprim, amino thiazole, and chloramphenicol. According to the instructions, 100 µl fresh-cultured *E. coli* O124:K72 D-09 (1×10^5 to 2×10^5 CFU/ml) was added into 96-well plates that contained different types and concentrations of antibiotics and incubated at 37 °C for 16–20 h. Bacterial growth or inhibition was observed in order to determine the MIC, which refers to the lowest antibiotic concentration that could inhibit the growth of the bacteria.

Animal model and strains

All animal experiments were performed in accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. The animal protocols were approved by the Committee on the Ethics of Animal Experiments (SYXK (Liao) 2018–0006). Forty 6-week-old male guinea pigs obtained from the Specific Pathogen Free Animal Center of Dalian Medical University were randomly divided into 4 groups. The animals were housed in groups with five guinea pigs per cage, and they were kept in a temperature-controlled room with free access to normal food and water for 1 week before experimentation.

Strains *E. coli* O124 K72 D-09 cultured in LB liquid medium. *Lactobacillus rhamnosus* GG (LGG) obtained

from Chinese General Microbiological Culture Collection (1.2134) was taken in negative control group and cultured in Lactobacilli MRS Broth.

Grouping Group 1 (G1) was the blank control group. With regard to the bacteria-treated groups, 1×10^8 CFU *E. coli* D-09 were intragastrically administered daily (G3), while the same amount of LGG was inoculated (G2) as a negative control group; the last group received a high cholesterol diet (2% cholesterol) (G4) as a positive control group. After 30 days, all animals were anesthetized by intraperitoneal injection of 10% chloral hydrate (0.3 ml/100 g body weight); blood samples were collected from the heart. Colon and liver samples were collected and divided into two parts; some were fixed in 4% paraformaldehyde for immunofluorescence analysis, and the others were immediately frozen and stored at -80°C for western blotting analysis.

Plasma diamine oxidase measurement

The plasma levels of diamine oxidase (DAO) in the different groups were measured via a DAO quantitation kit for guinea pigs. According to the instructions, the reaction between DAO and the substrate produces a chromogenic substance that can be detected at a wavelength of 450 nm in a microplate reader (Thermo, USA). The DAO level is directly proportional to the absorbance value (OD value); therefore, the levels of plasma DAO can be calculated.

Immunofluorescence analysis of claudin-2 and occludin expression

Colon and liver tissue were freshly dissected and immersed in 4% paraformaldehyde fixation for more than 12 h. After dehydration in 30% sucrose solution and embedding in optimal cutting temperature compound (SAKURA, USA), frozen sections were cut, and the prepared slides were stored at -20°C for immunofluorescence analysis. The immunofluorescence analysis was carried out according to Zhang's research with little alteration (Zhang et al. 2013). The slides were incubated overnight at 4°C with primary antibodies: rabbit anti-claudin-2 and rabbit anti-occludin (1:200, Abcam, USA) and then incubated for 1 h at 37°C with secondary antibodies: Alexa-488-labeled anti-rabbit IgG (1:1000, Jackson ImmunoResearch Laboratories, USA). With fluorescent mounting media, specimens were examined and photographed through a BX16 fluorescent microscope (Olympus, Japan).

Western blotting assay

The protein of proximal colon and liver was extracted via Total Protein Extraction Kit (Keygene, China), and protein

concentrations were determined using a BCA protein assay kit. Automated western blotting (ProteinSimple Wes, USA) was adopted to quantitatively analyze the changes in claudin-2 and occludin expression among groups. Simple western analyses were performed according to the ProteinSimple user manual. Protein (10 μg) was loaded, and the dilution ratio of primary antibody claudin-2 and occludin (Abcam, USA) was 1:1000.

Genome sequencing and annotation

DNA library construction was performed following the manufacturer's instructions (Illumina HiSeq 2500). One paired-end library with an insert size of 350–450 bp for each sample was built and sequenced with 125 bp read length from each end. The samples were sequenced using the Illumina HiSeq 2500 Platform. The following criteria were used for quality control: all the datasets were preprocessed as FASTQ short-reads using the FASTX-Toolkit software; sequence analysis of bacterial whole genomes was performed; the short scaffolds were filtered using a minimum-length threshold of 200 bp. For each genome, genes were predicted with Prodigal (Hyatt et al. 2010), and the antibiotic resistance genes were annotated by searching the ARDB (Liu and Pop 2009) and NCBI-NR databases using BLASTP.

Statistical analysis

Statistical significance was determined using ANOVA by SPSS (SPSS Statistics 17). Statistical histograms were performed by GraphPad Prism 6 (GraphPad Software, La Jolla, CA, USA). Data are expressed as the mean $\pm$ SD. *p* values < 0.05 were considered statistically significant.

Results

Identification of bacterial strain and MIC testing

The serotype of the pathogenic *E. coli* strain was identified as O124:K72, which belongs to EIEC (enteroinvasive *E. coli*) (Liu et al. 2015). The results of MIC showed that *E. coli* D-09 was sensitive to most of the antibiotics, except for gentamicin. Table 1 shows this information in detail.

E. coli O124:K72 D-09 infection destroys intestinal barrier function

As one type of enzyme secreted by intestinal mucosal cells, serum DAO levels are a good marker to reflect the integrity of the intestinal mucosa (Su et al. 2009). In this study, the level of plasma DAO was increased in the *E. coli* group and the high-cholesterol diet group ($p < 0.05$)

Table 1 Drug resistance analysis of *E. coli* D-09

Antibiotic	MIC (μg/ml)	Resistant/ sensitive (R/S)
Tetracycline	0.0625	S
Ciprofloxacin	2	S
Nalidixic acid	0.5	S
Cefoxitin	2	S
Cefotaxime	0.0625	S
Gentamicin	–	R
Trimethoprim/sulfonamide	0.0625/1.1875	S
Amine thiazole	0.0625/1.1875	S
Chloramphenicol	8	S

DAO concentration in blood serum

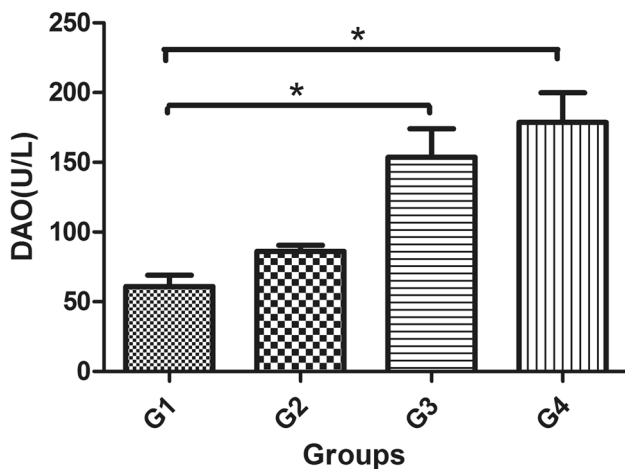


Fig. 1 Serum DAO levels were different between each group (G1: normal control group; G2: probiotic group; G3: *E. coli* group; G4: High cholesterol diet group). * $p < 0.05$ versus control group (G1)

compared with that in the control group, which suggested injury to the intestinal mucosa. In the probiotic group, although the level of plasma DAO was slightly higher than that of the control group, this difference was not statistically significant (Fig. 1). These changes revealed that *E. coli* D-09 could destroy the integrity of intestinal mucosa, causing an increase in intestinal permeability, similar to a high-cholesterol diet. However, the probiotics did not have this destructive effect. Wang's work established a NASH (non-alcoholic steatohepatitis) rat model by feeding rats a high-fat diet and found that endotoxin and DAO levels, which are markers of intestinal mucosal barrier function, were significantly higher in the high-fat diet group than in the control group (Wang et al. 2015).

Tight junctions (TJs) seal the space between adjacent epithelial cells. Mounting evidence suggests that tight junction proteins play a key role in the pathogenesis of human disease. Defects in the functionality of different protein components of tight junction such as occludins, claudins, E-cadherin, and ZO-1 are responsible for the pathogenesis of several human diseases (Chiba et al. 2008). Among them, claudin-2 is considered as a leaky protein and its overexpression associated with enhanced cell permeability (Zhang et al. 2013). Occludin, another type of TJ protein, is known to tighten the epithelial structure and exhibits ubiquitous expression under physiological conditions (Krug et al. 2014). In this research, immunofluorescence analysis of claudin-2 and occludin expression both in the colon and liver tissue of the *E. coli* group (G3) and the control group (G1) was done, and western blotting was employed to quantitatively analyze the changes in claudin-2 and occludin expression among groups.

In the colon, the results showed that claudin-2 expression along the crypt-villus axis of the intestinal epithelial cells was obviously increased in G3 group compared with G1 (Fig. 2a). In contrast, occludin expression was slightly decreased in the colon of G3 group, but without being significantly different. However, the intestinal villus was disordered, and some crevices between the intestinal villi could be observed in some parts of the intestinal segments in G3 group (Fig. 2b), revealing the damaged integrity of the intestinal barrier and increased cell permeability in the colon after *E. coli* O124:K72 invasion.

In the liver, claudin-2 expression in the connections between liver cells was also increased significantly in G3 group, especially near the edge of the liver blood vessel zone (Fig. 2c) while occludin expression was decreased in G3 group compared with G1 group (Fig. 2d).

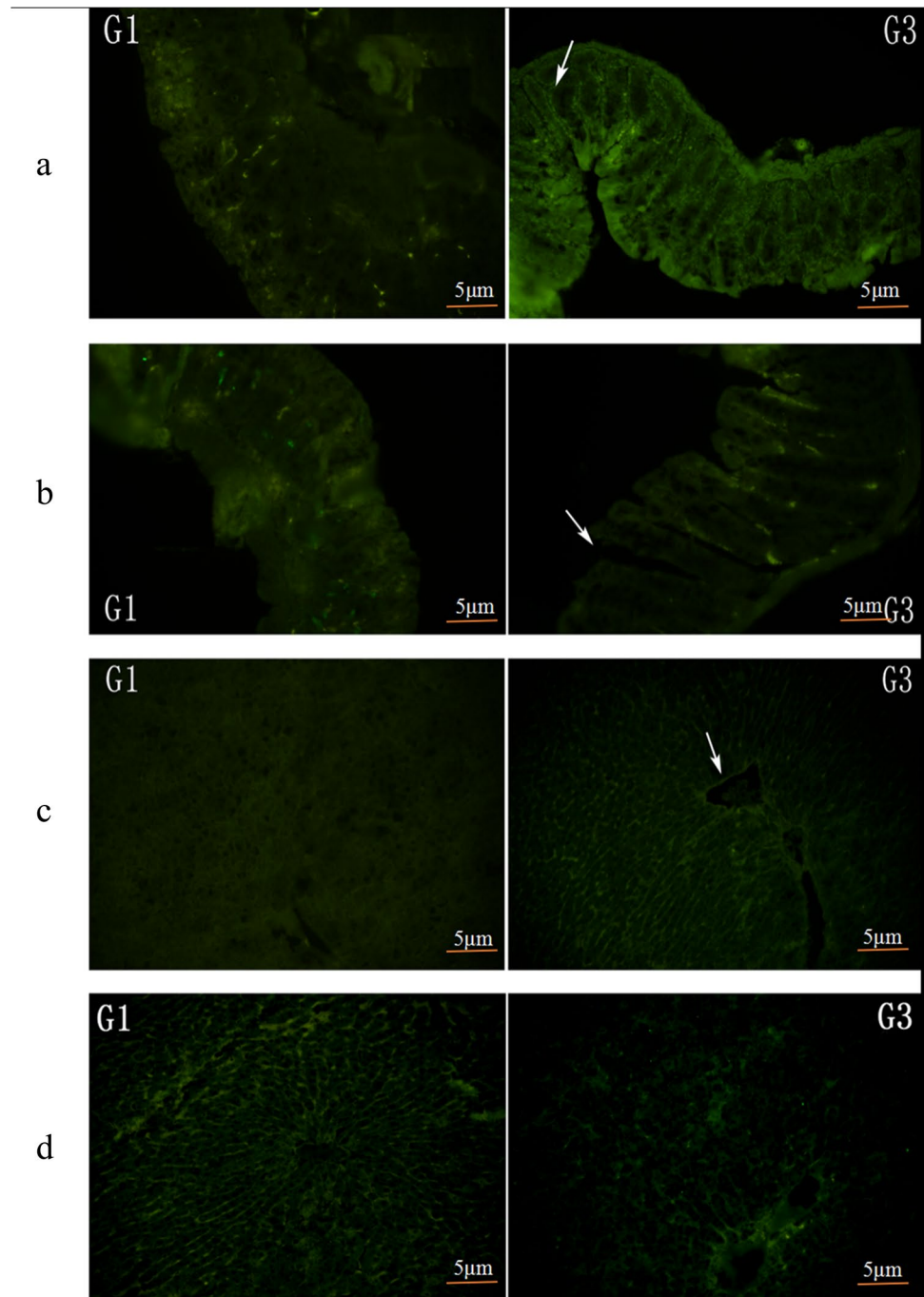
According to the results of western blotting, comparing to G1, the protein level of occludin was significantly upregulated in G2 and downregulated in G4 in liver (Fig. 3a, c). The protein level of claudin-2 was markedly increased both in liver and colon in G3 (Fig. 3b, d, and e). The occludin expression was not significantly different in colon among groups.

Therefore, the upregulated expression of claudin-2 both in the colon and in the liver might have caused damage to intestinal and liver barrier function in the *E. coli* group.

Genomic characteristics of *E. coli* O124:K72 D-09

The assembled draft genome of *E. coli* D-09 comprises 42 scaffolds (size, > 645 bp) with a total size of 4,679,567 bp, an N50 contig length of 222,736 nucleotides, and a mean G + C content of 50.69%. A total of 4379 protein coding genes covering 88.0% of the genome, 80 tRNAs (representing all 20 amino acids), and 5 rRNAs (5S,

Fig. 2 Immunofluorescence detection of the expression of claudin-2 (**a**) and occludin (**b**) in the colon between G1 and G3 groups. Claudin-2 was upregulated in G3 compared with G1; the expression of occludin slightly decreased in G3, and irregularly arranged intestinal villi could be observed in this group (arrow). Immunofluorescence detection of the expression of claudin-2 (**c**) and occludin (**d**) in the liver of the G1 and G3 groups. Claudin-2 was upregulated in G3 compared with G1, especially near the edge of blood vessels (arrow); The expression of occludin was downregulated in the G3 group



16S, and 23S) were annotated. Comparison of the D-09 draft genome with all previously sequenced genomes of *E. coli* in NCBI database isolates revealed a very close relationship to *E. coli* K-12 strain, with an average nucleotide identity (ANI) of 99.97% to K-12 sub-strains W3110 and MG1655 and 99.96% to DH10B and MDS42. Therefore, these top four sub-strains of *E. coli* K-12 were chosen for comparative genomic analysis (Fig. 4), including W3110, MG1655, DH10B, and MDS42 with the genome accession number of NC 007779.1,

NC000913.3, NC010473.1, and NC020518.1, respectively in NCBI database. The D-09 genome contains 5 antibiotic resistance genes and 40 multidrug resistance genes (Table S1). All of the antibiotic resistance genes, including *ampC* (β -lactamase), *folA* (dihydrofolate reductase), *folP* (dihydropteroate synthase), *cmr* (chloramphenicol resistance), and *tet* (B) (tetracycline resistance), refer to low level natural resistance to corresponding antibiotics. Eighty-six virulence-related genes were also identified in the D-09 genome, including the ferric enterobactin

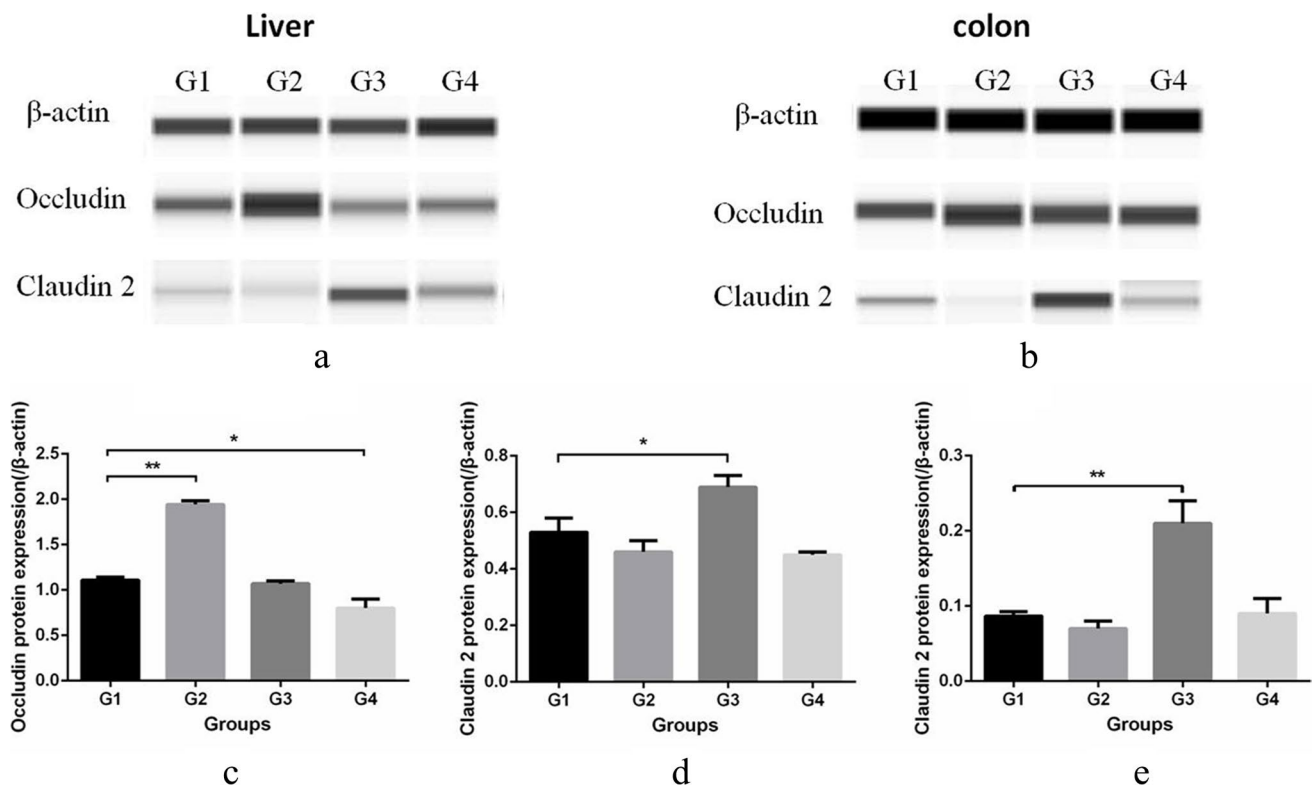


Fig. 3 Expression of occludin and claudin-2 in liver (**a**) and colon (**b**) based on the western blotting assay. Statistical comparisons of the occludin expressions levels in liver (**c**), as well as of the claudin-2

expressions in liver (**d**) and colon (**e**) (G1: Normal control group; G2: Probiotic group; G3: *E. coli* group; G4: high cholesterol diet group). * $p < 0.05$; ** $p < 0.01$ versus control group (G1)

synthesis and transport system and flagella and fimbriae biosynthetic proteins (Table S2).

The carbohydrates of pathogens may be associated with the adhesion and invasion process. Carbohydrate-Active enZymes Database (CAZy), a professional database related to carbohydrate enzymes, contains enzyme families that can catalyze carbohydrate degradation, modification, and biosynthesis (Cantarel et al. 2009). Bacteria closely related to human health and disease encode hundreds of CAZyme genes in their genomes for degrading various dietary and host carbohydrates (Huang et al. 2018), which is very important for the growth, reproduction, and invasion of bacteria. Therefore, we compared the CAZymes of *E. coli* D-09 with other four sub-strains of *Escherichia coli* K-12 to investigate whether *E. coli* D-09 contained specific enzymes by analyzing in the CAZy database (<http://www.cazy.org/>). Some genes involved in the regulation of carbohydrates only existed in the *E. coli* D-09 genome, including 0001 (transcriptional regulator, ARAC family), 1189 (transglycosylase), 3779 (chaperone), 3785 (mannonate dehydratase), and 3793 [beta-aspartyl-dipeptidase (metallo-type)]. Figure 5 shows this information in detail.

Discussion

It was not completely consistent between MIC test results and the mechanisms present in the genome sequence, such as the resistance determining genes *folA/folP* or *tetB*. *FolA* (dihydrofolate reductase) and *folP* (dihydropterolate synthase) are a kind of natural resistance against sulfamethoxazole-trimethoprim (SMZ-TMP), which can be resistant to low concentration of SMZ-TMP (Rolain and Raoult 2005). Due to the low drug-resistance, the bacteria that only contain *folA/P* usually show the resistance to SMZ-TMP. Similarly, tetracycline resistance also involves in complex mechanism, such as ribosomal RNA methylation, efflux, drug modification, specific gene mutations, and the employment of specialized ribosome protection proteins (Nguyen et al. 2014). Therefore, the drug resistance of bacteria is not only determined by drug resistance genes, but also regulated by other factors. *TetB* belongs to low-level resistant gene. That means the bacteria containing this gene usually show low-level resistance to tetracycline (Sloan et al. 1994). However, it is also known that some bacteria containing *tetB* gene are sensitive to tetracycline (Park

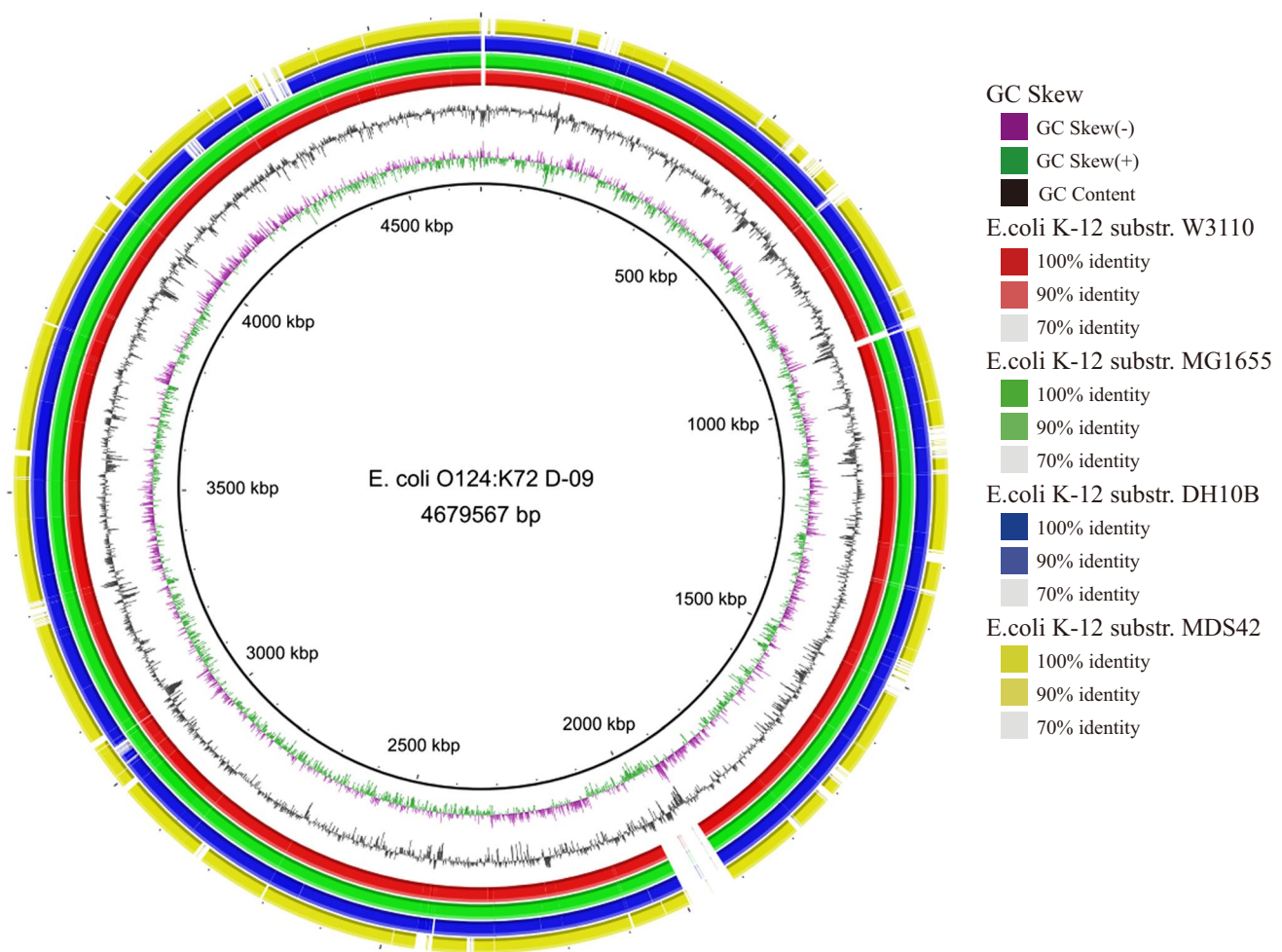


Fig. 4 Circular representation of the *E. coli* O124:K72 D-09 genome and its comparison with K-12 sub-strains. The inner three circles represent the genome scale, GC skew, and GC content of the

D-09 genome, and the outer four circles show the regions of the *E. coli* D-09 region that are closely orthologous with K-12 sub-strains W3110 (red), MG1655 (green), DH10B (blue), and MDS42 (yellow)

et al. 2010). Therefore, in clinical practice, the bacterial resistance is subject to drug-sensitive test.

EIEC could impair intestinal barrier integrity through affecting the expression of TJ proteins, which facilitate the bacteria to penetrate the intestinal barrier and cause inflammation (Pawlowska and Sobieszczanska 2017). Claudin-2 is a type of epithelial tight junction protein and exhibits higher expression in leaky epithelia; it is the so-called leaky protein (Luettiga et al. 2015). The expression of claudin-2 is changed under different pathologic conditions, such as Crohn's disease (Zeissig et al. 2007), ulcerative colitis (Heller et al. 2005), and Celiac disease (Schumann et al. 2012a, 2012b). Overexpression of claudin-2 resulted in an increase in epithelial barrier permeability both in vitro and in vivo (Zeissig et al. 2007; Tsai et al. 2017). Occludin was also an important TJ protein. The downregulated expression of occludin has been observed in inflammatory diseases (Zeissig et al. 2007; Markov et al. 2011). This research

found that claudin-2 expression increased obviously in the colon and liver of guinea pigs infected with *E. coli* D-09. The serum levels of diamine oxidase were significantly increased, illustrating a higher intestinal permeability. Toward occludin, we found the protein level of occludin was significantly upregulated in the liver of the LGG group and downregulated in the high-cholesterol diet group while there were no significant differences in colon among groups. The data indicated that *E. coli* D-09 was likely to increase intestinal permeability through upregulating claudin-2 expression between intestinal epithelial cells. The increased intestinal permeability facilitates bacterial and endotoxin translocation. This result was consistent with Ren's report, and they also found that this bacterium could influence the formation and secretion of mucin in the intestine, such as MUC2, MUC1, MUC13, and MUC15 (Ren et al. 2017). While the mechanism of high cholesterol diet causing the increased intestinal permeability might not relate to the

CAZy type	O124:K 72 D-09	DH10B	MDS42	MG1655	W3110	Description
GH39	exist	non-exist	exist	exist	exist	Transcriptional regulator, ARAC family
GT2	exist	exist	exist	exist	exist	iron complex transport system ATP-binding protein
CEnc	exist	exist	exist	exist	exist	Retrotransposon protein
GH24	exist	exist	exist	exist	exist	lysozyme
GT2	exist	exist	exist	exist	exist	Glycosyl transferase, family 2
CE14	exist	exist	exist	exist	exist	ImbE family
GT4	exist	exist	exist	exist	exist	taurine-transporting ATPase
GT2	exist	exist	exist	exist	exist	none
GH24	exist	exist	exist	exist	exist	lysozyme
GH13	exist	exist	exist	exist	exist	alpha-amylase
GH23	exist	exist	exist	exist	exist	(ABC) transporter
GT2	exist	exist	exist	exist	exist	response regulator
GH2	exist	exist	exist	exist	exist	homocysteine S-methyltransferase
GH23	exist	exist	exist	exist	exist	transglycosylase
GH73	exist	exist	exist	exist	exist	flagellar rod assembly protein muramidase flgJ
CBM5	exist	exist	exist	exist	exist	bifunctional chitinase/lysozyme
GH13	exist	exist	exist	exist	exist	alpha amylase, catalytic
GH13	exist	exist	exist	exist	exist	alpha amylase, catalytic
GH13	exist	exist	exist	exist	exist	alpha amylase, catalytic
GH43	exist	exist	exist	exist	exist	xylan 1,4-beta-xylosidase
GH43	exist	exist	exist	exist	exist	Transporter
GH33	exist	exist	exist	exist	exist	dihydrodipicolinate synthase
GT2	exist	exist	exist	exist	exist	dolichol-phosphate mannosyltransferase
GT4	exist	exist	exist	exist	exist	iron complex transport system ATP-binding protein
GT4	exist	exist	exist	exist	exist	(ABC) transporter
GT2	exist	exist	exist	exist	exist	biofilm PGA synthesis N-glycosyltransferase PgaC
CE4	exist	exist	exist	exist	exist	biofilm PGA synthesis lipoprotein PgaB
GT8	exist	exist	exist	exist	exist	Chaperone
GH35	exist	exist	exist	exist	exist	mannonate dehydratase
CE9	exist	exist	exist	exist	exist	beta-aspartyl-dipeptidase (metallo-type)
GT2	exist	exist	exist	exist	exist	Glycosyl transferase, family 2
GT2	exist	exist	exist	exist	exist	Glycosyl transferase, family 2
GT4	exist	exist	exist	exist	exist	Glycosyl transferase (Group 1)
GT4	exist	exist	exist	exist	exist	UDP-galactopyranose mutase
GT2	exist	exist	exist	exist	exist	dTDP-4-dehydrorhamnose reductase
GT4	exist	exist	exist	exist	exist	Glycosyl transferase (Group 1)
GT4	exist	exist	exist	exist	exist	Polysaccharide biosynthesis protein
GT4	exist	exist	exist	exist	exist	Glycosyl transferase (Group 1)
GT2	exist	exist	exist	exist	exist	putative colanic acid biosynthesis glycosyltransferase
GT4	exist	exist	exist	exist	exist	putative colanic acid biosynthesis glycosyltransferase
GT2	exist	exist	exist	exist	exist	putative colanic acid biosynthesis acetyltransferase WcaB
GT2	exist	exist	exist	exist	exist	Glycosyl transferase, family 2



Fig. 5 The carbohydrate-related genes differ between the genomes of *E. coli* O124:K72 D-09 and the K-12 sub-strains (DH10B, MDS 42, MG1655, and W3110)

claudin 2. It was reported that claudin-2 could serve as a gateway for adherent-invasive *Escherichia coli* (AIEC) entry into intestinal epithelial cells while co-internalization of TJ proteins and AIEC may promote AIEC invasion and increased barrier permeability (Shawki and McCole 2017).

According to *E. coli* D-09 whole-genome sequencing, this strain contains plenty of flagella and fimbriae biosynthetic genes. They are not only the motor organs but also virulence factors that facilitate the adhesion to and invasion of host cells, which are closely related to the pathogenicity of bacteria. Therefore, the invasion of *E. coli* D-09 may be associated with the fimbriae. For example, it contains an intact type 1 pilus gene cluster. The identified function of type 1 pili is to mediate adherence to host cells. Denizot's research found that AIEC exists widely in the ileum of Crohn's disease patients (Denizot et al. 2012). AIEC can upregulate the expression of claudin-2 and increase intestinal permeability in mice via type 1 pili. In this research, upregulated claudin-2 expression and increased intestinal permeability were also observed in guinea pigs infected with *E. coli* D-09. It needs further study to investigate whether its molecular mechanism is related to type 1 pili.

According to the whole-genome sequencing analysis, the *E. coli* D-09 genome contains four types of carbohydrate-active enzymes, which differ from those in the *E. coli* K-12 genome. These genes were a transcriptional regulator, transglycosylase, mannonate dehydratase, and beta-aspartyl-dipeptidase (metallo-type). According to some studies, transglycosylase participates in the synthesis of capsular polysaccharide (CPS), which is an essential virulence factor for some pathogens to infect the host (Zhang et al. 2016). The genes that encode glucosyltransferases (gtfs), especially gtfB and gtfC, are important in *Streptococcus mutans* colonization and pathogenesis (Yousefi et al. 2012). Mannose dehydratase (ManD), an enzyme found among bacterial species, dehydrates d-mannose to form 2-keto-3-deoxygluconate (2-KDG) as carbon and energy source. The drug targeting this enzyme might be of great therapeutic potential to treat *Streptococcus suis* infections (Zhang et al. 2009). Nevertheless, the functions of these four types of carbohydrate-active enzymes in *E. coli* D-09 need further study.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10123-022-00255-8>.

Author contribution D.S., Y.X., and J.J.L. designed all the experiments. J.J.L., J.J.Q., X.S., and X.Q.S. performed the experiments. J.J.Q., F.B.S., and J.Y.J. did the statistics. J.J.L. wrote the manuscript. D.S. and Y.X. corrected the manuscript. A.A. edited the English language. All the authors read and approved the final manuscript.

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Data availability The raw data of bacterial whole-genome sequencing in this study have been deposited in the Sequence Read Archive (SRA) in NCBI database with the accession number SRP093203: PRJNA353183 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA353183>).

Declarations

Ethics approval and consent to participate All animal experiments were performed in accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. The animal protocols were approved by the Committee on the Ethics of Animal Experiments of the Dalian Medical University.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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